

Final Exam, math54.
Version A.

Instructions. Do not forget to mark your version on the bubble sheet. **This version is A.** Every question has only one correct choice. You should fill in only one bubble for each question.

1. Let $A = \begin{bmatrix} 2 & h & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. The matrix A is diagonalizable if and only if

(A)	(B)	(C)	(D)	(E)
$h \neq 0$	$h = 0$	h is any real number	$h \neq 2$	$h = 2$

For questions 2 and 3 consider the vector space $\mathbb{P}_2$ of polynomials of degree ≤ 2 and the linear transformation $T : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ defined by the formula

$$T(f) = t^2 f'' + f' + f.$$

2. The matrix of T relative to the basis $\{1, t, t^2\}$ is

(A)	(B)	(C)	(D)	(E)
$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

3. The eigenvalues of T are

(A)	(B)	(C)	(D)	(E)
0, 1, 2	2, 4	1, 2, 3	0, 1, 3	1, 3

4. Find the closest point to $\begin{bmatrix} 3 \\ 5 \\ 4 \\ 1 \end{bmatrix}$ in the subspace W of $\mathbb{R}^4$ spanned by $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

(A)	(B)	(C)	(D)	(E)
$\begin{bmatrix} -1/2 \\ 5/2 \\ 5/2 \\ -1/2 \end{bmatrix}$	$\begin{bmatrix} -1/3 \\ 1/3 \\ 1/3 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 4/3 \\ -2/3 \\ -2/3 \\ 2 \end{bmatrix}$	$\begin{bmatrix} -1/3 \\ 1 \\ -2/3 \\ 1/3 \end{bmatrix}$	$\begin{bmatrix} 1/2 \\ 5/2 \\ 0 \\ -1/2 \end{bmatrix}$

5. Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the data

$$(-1, 2), (0, 5), (1, 6), (2, 9)$$

(A)	(B)	(C)	(D)	(E)
$y = -0.3 - 1.4x$	$y = 2.5 + 2x$	$y = 4.4 + 2.2x$	$y = 1.1 - x$	$y = 3.3 - 2.2x$

For questions 6 and 7 consider the quadratic form in $\mathbb{R}^2$ given by the formula

$$Q(\mathbf{x}) = x_1^2 + 2x_1x_2 + x_2^2.$$

6. The form $Q(\mathbf{x})$ can be written as $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ where the matrix A is

(A)	(B)	(C)	(D)	(E)
$\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

7. The quadratic form $Q(x)$ is

(A)	(B)	(C)	(D)	(E)
positive definite	negative definite	positive semidefinite	negative semidefinite	indefinite

8. To find a particular solution to the equation

$$y'' + 2y' + y = e^{-t}$$

one should use the form

(A)	(B)	(C)	(D)	(E)
Ate^t	Ae^{-t}	Ate^{-t}	At^2e^{-t}	At^3e^{-t}

9. The differential equation $y'' + 3y' + 2y = 0$ is equivalent to the system $\mathbf{x}'(t) = A\mathbf{x}(t)$ where A is equal to

(A)	(B)	(C)	(D)	(E)
$\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ -3 & -2 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$

10. Consider the linear system with $\mathbf{x}'(t) = A\mathbf{x}(t)$ with the fundamental matrix

$$X(t) = \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix}.$$

Let $\mathbf{x}(t)$ be the solution such that $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Then $\mathbf{x}(1)$ is equal to

(A)	(B)	(C)	(D)	(E)
$\begin{bmatrix} e \\ e^2 \end{bmatrix}$	$\begin{bmatrix} e \\ e \end{bmatrix}$	$\begin{bmatrix} e \\ 2e \end{bmatrix}$	$\begin{bmatrix} e \\ 0 \end{bmatrix}$	$\begin{bmatrix} 2e \\ e \end{bmatrix}$